

Tutorial of energy minimization for molecules by MolAICal

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1. Introduction

Sometimes, the docked ligands in the pocket of receptors appear slightly conformationally distorted, and some researchers need to optimize the structures of ligands for some purpose. The molecular energy minimization can provide a good way to optimize the molecules into more reasonable state. Here, MolAICal supplies two ways for molecular energy minimization: one is the minimization implementation by MolAICal, the other is based on the interface of MolAICal and UCSF Chimera. For more introduction to energy minimization for molecules, please see the manual of MolAICal.

2. Materials

2.1. Software requirements

- 1) MolAICal: <https://molaical.github.io> or <https://molaical.gitlab.io>
- 2) UCSF Chimera: <https://www.cgl.ucsf.edu/chimera>

Notice: Make sure the installation of MolAICal is correct!

2.2. Example files

- 1) All the necessary tutorial files are downloaded from:
<https://github.com/MolAICal/tutorials/tree/master/025-minimization>

This tutorial presents two distinct approaches for energy minimization, detailed in Part I and Part II, respectively. Readers may choose to consult **either section individually** or both sequentially, depending on their requirements.

Part I. Energy minimization by MolAICal

Go into the folder “[025-minimization](#)”, which contains some protein files (GCGRH.mol2 and GCGRH.pdb), ligand files (lig_10.mol2 and lig_10.pdb), and the files including the character “_fix” to assign the fixed atoms for energy minimization.

Check one or two molecule information for ensuring atom serials of the inputted molecules:

```
#> molaical.exe -min1 -i lig_10.mol2  
#> molaical.exe -min2 -m info -i1 GCGRH.pdb -i2 lig_10.pdb
```

For minimizing one molecule:

- 1) Minimize a single molecule without the fixed atoms

```
#> molaical.exe -min1 -i lig_10.mol2 -o min_lig.mol2
```
- 2) Minimize a single molecule with the serial number of fixed atoms

```
#> molaical.exe -min1 -i lig_10.mol2 -o min_lig.mol2 -f 1-2
```
- 3) Minimize a single molecule with the reference fixed molecule file, iteration numbers of quick optimization and refinement

```
#> molaical.exe -min1 -i lig_10.mol2 -o min_lig.mol2 -f lig_fix.mol2 -q 10 -r 6
```

Minimize two molecules, output only the result of the second input molecule:

1) Minimize two input molecules

```
#> molaical.exe -min2 -m 1 -i1 GCGRH.pdb -i2 lig_10.pdb -f1 GCGRH_fix.pdb -f2 0 -o2 min_lig_1.pdb
```

2) Convert the second input PDB molecule in Mol2 and minimize two input molecules

```
#> molaical.exe -min2 -m 1 -i1 GCGRH.pdb -i2 lig_10.pdb -f1 GCGRH_fix.pdb -f2 0 -o2 min_lig_2.pdb -c2
```

Analysis: If “-f2” is 0, it means no fixed atoms the second molecule. Open ligands: [lig_10.pdb](#), [min_lig_1.pdb](#), and [min_lig_2.pdb](#). Because the molecule in **PDB format has no bond connectivity information, some bonds will be changed** when energy minimization is performed. Please see Figure 1. **The molecule in mol2 format contains bond connection information, so it has the bond missing issue.** For fixed atoms, bond connectivity remains unchanged; thus, the fixed PDB-format molecules can be directly used for energy minimization without bond parameter consideration. “-c2” option converts the second molecule (in PDB format) to mol2 format and then performs energy minimization on the converted structure.

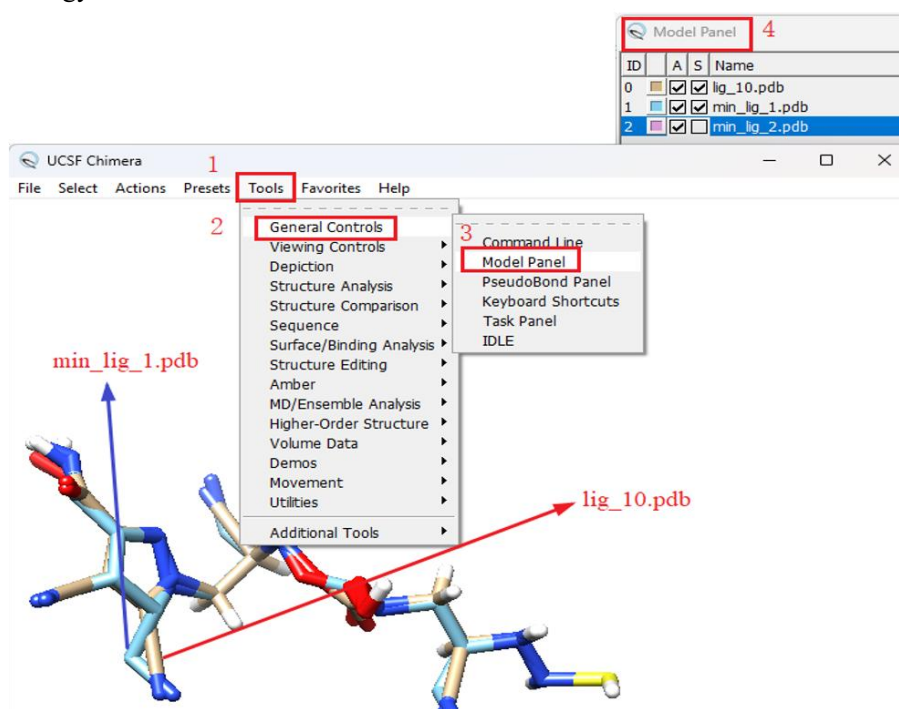


Figure 1

Minimize two molecules, output results for both input molecules:

1) Minimize two input molecules

```
#> molaical.exe -min2 -m 2 -i1 GCGRH.pdb -i2 lig_10.pdb -o1 min_gcgr.mol2 -o2 min_lig.mol2
```

2) Convert the second input PDB molecule in Mol2 and minimize two input molecules

```
#> molaical.exe -min2 -m 2 -i1 GCGRH.pdb -i2 lig_10.pdb -o1 min_gcgr.mol2 -o2 min_lig.mol2 -c2
```

For the parameters of minimization of MolAICal, please see the content below:

-f1 <arg>	Either fixed atoms (e.g., '1-3,5,7-9') or reference molecule file for first molecule.
-f2 <arg>	Either fixed atoms (e.g., '1-3,5,7-9') or reference molecule file for second molecule.
-h	Print help information.
-i1 <arg>	Input molecule file for first molecule (.pdb, .mol2, .sdf, .mol).
-i2 <arg>	Input molecule file for second molecule (.pdb, .mol2, .sdf, .mol).
-m <arg>	Operation mode for minimization (default: null), value can be info, 1, or 2 : “ info ” means to print molecules' information only; “ 1 ” means to minimize one molecule only; “ 2 ” means to minimize two molecules.
-o1 <arg>	Output file for minimized molecule for first molecule (.pdb, .mol2, .sdf, .mol).
-o2 <arg>	Output file for minimized molecule for second molecule (.pdb, .mol2, .sdf, .mol).
-q <arg>	The iteration number of Quick optimization.
-r <arg>	The iteration number of Refinement.
-c2	Convert the second input PDB file in mol2 format if "-c" in the command line, it is designed for molecular docking of MolAICal.

Part II. Energy minimization by UCSF Chimera and MolAICal

Should energy minimization in Part I suffice for the research objectives, this section may be omitted. In cases where the Part I procedure yields erroneous results, Part II serves as a validated alternative methodology. Go into the folder “[025-minimization](#)”:

Mode 0: It can run the script file of UCSF Chimera directly

1) Run the command script of UCSF Chimera by MolAICal

```
#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 0 -i1 run.com
```

Note: “-c” is needed to point to the path of the program “chimera”, “run.com” is the command script file of UCSF Chimera.

Mode 1: minimization for one molecule

1) Minimize one molecule with the assigned atom serials

```
#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 1 -i1 lig_10.mol2 -sv "#0:@/serialNumber>=20" -o1 out_1.mol2
```

Note: Here, the option “-i1” is the inputted molecular file, which is different from Mode 0. The option “-sv” points to the atom specification of UCSF Chimera, please see https://www.rbvi.ucsf.edu/chimera/current/docs/UsersGuide/midas/frameatom_spec.html)

Mode 2: minimization for two molecules

1) Minimize two molecules with the assigned atom specifications and output two molecules

```
#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 2 -i1 lig_10.mol2 -i2 GCGRH.pdb -sv "#1" -os "#2:<0>" -o1 out_1.mol2 -o2 out_2.pdb
```

Note: the options “-sv” and “-os” point to the atom specification of UCSF Chimera.

Mode 3: minimization for two molecules with one fixed molecule

1) Minimize two molecules with one fixed molecule and output two molecules

```
#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 3 -i1 lig_10.mol2 -i2 GCGRH.pdb -f1 GCGRH_fix.pdb -os "#3:<0>" -o1 out_1.mol2 -o2 out_2.pdb
```

Mode 4: minimization for two molecules with two fixed molecules

1) Minimize two molecules with two fixed molecules, output one molecule and log file based on the full options of MolAICal

```
#> molaical.exe -cmin -c "/home/feng/soft/chimera115/bin/chimera" -m 4 -i1 lig_10.mol2 -i2 GCGRH.pdb -f1 GCGRH_fix.pdb -f2 lig_fix.mol2 -os "#4:<0>" -cm gas -fs freeze -sv "#0,1 za<0.01" -fv false -pv false -ns 10 -ss 0.02 -cs 2 -css 0.02 -iv 1 -ng true -mf1 mol2 -mf2 pdb -o1 out_1.mol2 -o2 out_2.pdb -w false -p false
```

For the parameters of minimization of UCSF Chimera and MolAICal, please see the content below:

-c,--chimera_path <arg>	Path to Chimera executable
-cm,--charge_method <arg>	Charge method (am1 or gas), default: gas
-cmin	minimization calculation from the interface for UCSF Chimera.
-cs,--cgsteps_value <arg>	Conjugate Gradient (CG) steps, default: 2
-css,--cgstepsize_value <arg>	Conjugate Gradient (CG) step size, default: 0.02
-f1,--fixed_f1 <arg>	Path to first fixed molecule file (for modes 3 and 4)
-f2,--fixed_f2 <arg>	Path to second fixed molecule file (for mode 4)
-fs,--freeze_spec <arg>	Freeze or spec method, default: freeze
-fv,--fragment_value <arg>	Whether to use fragment (true/false), default: false
-h,--help	Display help information
-i1,--input_f1 <arg>	Path to first input molecule file or command file (for mode 0)
-i2,--input_f2 <arg>	Path to second input molecule file or output log (for mode 0)
-iv,--interval_value <arg>	Interval, default: 1
-m,--mode <arg>	Mode of operation (0, 1, 2, 3, or 4): Mode 0: Direct call to chimera_command (--input_f1 as commandFile, --input_f2 as outputLog); Mode 1: Single molecule minimization; Mode 2: Two-molecule minimization;

	Mode 3: Three-molecule minimization with one fixed;
	Mode 4: Four-molecule minimization with two fixed.
-mf1,--mol_format_1 <arg>	First output format, default: mol2
-mf2,--mol_format_2 <arg>	Second output format, default: pdb
-ng,--nogui_value <arg>	Whether to use GUI (true/false), default: true
-ns,--nsteps_value <arg>	Steepest descent steps, default: 10
-o1,--output_f1 <arg>	First output file
-o2,--output_f2 <arg>	Second output file
-os,--output_sel <arg>	Output selection
-p,--print_output <arg>	Whether to print output (true/false), false means to output the log file, vice versa, default: true
-pv,--prep_value <arg>	Whether to use prep (true/false), default: true
-ss,--stepsize_value <arg>	Steepest descent step size, default: 0.02
-sv,--sel_value <arg>	Selection value for atoms
-w,--whether_output <arg>	Whether to output the second molecule (true/false), false means no second output, vice versa, default: true